Blessed G/Michael Catholic Secondary School
G-12 Physics First Semester Final Exam

Topic: unit 1 up to unit 4 **Date: 18/05/2018 EC** **Time allowed: 1:30 hr**

Take g = 10m/s² density of water = 1000 kg/m³ Hanna Agalew

I. For each of the following questions, choose the best answer from the given alternatives.

A 1. The ability to control exhalation allows long phrases to be sung in a controlled and relaxed manner.
A) True B) False

B 2. Hyper-echoic areas appear black on ultrasound because they do not send back any sound waves.
A) True B) False

B 3. The direction of angular acceleration is in the same direction as the direction of angular velocity if the angular speed is decreasing in time. A) True B) False

A 4. When a wheel of radius *R* rotates about a fixed axis, all points on the wheel have the same angular speed. A) True B) False

A 5. According to Kepler's laws of planetary motion, a satellite increases its speed as it approaches the Sun and decreases its speed as it moves away from the Sun. A) True B) False

B 6. The gravitational force between a 60 kg man and Earth is not equal because Earth is more massive than the man therefore, it exerts the greatest force. A) True B) False

A 7. A sealed empty plastic water bottle collapses as an airplane descends from a higher elevation to a lower elevation. A) True B) False

A 8. As a fluid's speed increases, the pressure that the moving fluid exerts on the surface decreases.
A) True B) False

B 9. The magnetic field B at a distance r from a long straight wire carrying current I is directly proportional to r. A. True B. False

B 10. In Lenz's law the induced emf opposes the change in magnetic flux. A. True B. False

A 11. An electromagnet loses its magnetism when the electric current is off. A) True B) False

A 12. The magnetic field lines are denser where the magnetic field is stronger. A) True B) False

B 13. A rectangular wooden block floats with 75% of its volume inside water. What is the density of this block?
A. 75000 kg/m³ B. 750 kg/m³ C. 25000 kg/m³ D. 250 kg/m³

C 14. How would the period of an object in a circular orbit change if the radius of the orbit doubled?
A. The period would increase by a factor of 2.
B. The period would decrease by a factor of 4.
C. The period would increase by a factor of $2\sqrt{2}$.
D. The period would decrease by a factor of $2\sqrt{2}$.

$\frac{T^2}{R^3} = K \quad T = \sqrt{R^3 K}$

$\frac{T^2}{8R^3} = K \quad T = \sqrt{8}\sqrt{R^3 K}$

$\cos 30° = \sin 60° = \frac{\sqrt{3}}{2}, \cos 37° = \sin 53° = 0.8, \cos 45° = \sin 45° = \frac{\sqrt{2}}{2}, \cos 53° = \sin 37° = 0.6, \cos 60° = \sin 30° = 0.5$

C 15. If you double the mass of a certain substance, what will happen to its density?

A. It will be doubled.
C. It remains the same.
B. It will be halved.
D. It is indeterminate.

B 16. A fluid that occupies a volume of 20 L weighs 240 N at a location where the gravitational acceleration is 10 m/s^2. What is the mass of this fluid?

A. 20 kg.
B. 24 kg
C. 40 kg
D. 220 kg

D 17. The figure below shows an incompressible fluid flowing along a pipe of decreasing radius. Which one of the following is necessarily true?



A. The flow rate must be increasing along the pipe of decreasing radius.
B. The flow rate must be decreasing along the pipe of decreasing radius.
C. The fluid pressure must be increasing along the pipe of decreasing radius.
D. The fluid pressure must be decreasing along the pipe of decreasing radius.

D 18. The gauge pressure in a liquid at a depth of 2.5 m is read to be 28 kPa. What is the gauge pressure in the same liquid at a depth of 10 m?

A. 14 kPa
B. 84 kPa
C. 56 kPa
D. 112 kPa

C 19. Two identical balls A and B are kicked at an angle of 37^0 and 53^0 with the horizontal respectively, with the same initial velocity. Which ball has the maximum horizontal displacement?

A. Ball A
C. Both balls will have the same maximum horizontal displacement.
B. Ball B
D. Impossible to determine.

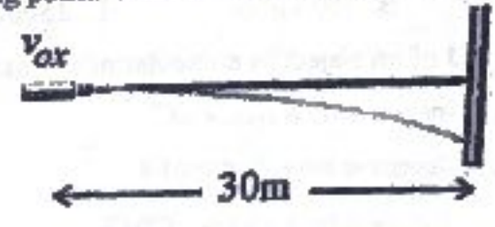
C 20. A projectile is fired in such a way that its horizontal range is equal to three times its maximum height. What is the angle of projection?

A. 30^0
B. 37^0
C. 53^0
D. 60^0

B 21. Which motion **CANNOT** an example of projectile motion?

A. A bullet fired from a gun.
C. A ball thrown horizontally into the air.
B. A bird flying in the air.
D. A javelin thrown by an athlete.

A 22. A rifle is aimed horizontally at a target 30m away as shown in figure below. The bullet hits the target 1.8 cm below the aiming point. What is the initial velocity of the bullet?



A. 500 m/s
B. 50 m/s
C. 600 m/s
D. 60 m/s

C 23. Lenz's law is the result of the law of conservation of:

A. mass
B. charge
C. energy
D. Momentum

D 24. A rescue airplane travelling at 360 km/h horizontally drops a food package from a height of 320 m when it passes over a car driver stranded in the desert. How far from the car driver should the food package be dropped?

A. 340 m B. 400 m C. 680 m D. 800 m

C 25. At which position in its flight will a ball experience its minimum speed during inclined projection?

A. at the beginning C. at maximum height

B. at the end D. the same speed at all positions

A 26. A transformer has primary coil with 2000 loops and secondary coil with 1000 loops. If the current in the primary coil is 4 A, then what is the current in the secondary coil?

A. 8 A B. 16 A C. 2 A D. 1 A

A 27. A wheel has a radius of 20 cm and accelerates from rest to 15 rev/s in 30 s. What is the magnitude of the tangential acceleration of a point at the tip of the wheel?

A. 0.6 m/s^2 B. 0.5 m/s^2 C. 2.0 m/s^2 D. 1.2 m/s^2

B 28. A ball is kicked into the air from the ground at an angle θ with the horizontal. When the ball reaches its highest point, which statement is true?

A. Both the velocity and acceleration of the ball are zero.

B. Its velocity is perpendicular to its acceleration.

C. Its velocity is not zero, but its acceleration is zero.

D. Its acceleration depends on the angle at which the ball was thrown.

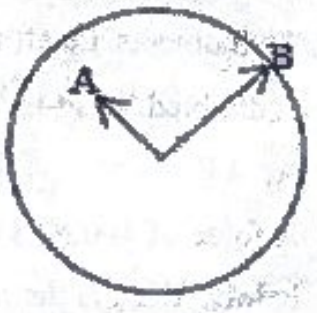
B 29. A ball is kicked from the ground with an initial speed of 25 m/s at an angle of 53^0 above the horizontal directly toward a wall. The wall is 24m from the release point of the ball. How long does the ball take to reach the wall?

A. 3.2 s B. 1.6 s C. 0.625 s D. 1.3 s

D 30. Which of the following **cannot** be a unit for angular displacement?

A. degree B. radian C. revolution D. meter

C 31. Consider two particles, A and B, on a flat rotating disk as shown in figure below. Particle A is 20 cm and particle B is 40 cm from the center. The disc starts from rest and its angular speed increases to 20 rad/s in 4s. What is the average angular and linear acceleration for particle B, respectively?



A. 5 rad/s^2, 1 m/s^2 C. 5 rad/s^2, 2 m/s^2

B. 10 rad/s^2, 2 m/s^2 D. 10 rad/s^2, 1 m/s^2

A 32. The magnetic flux is maximum when the angle between magnetic field lines and the line perpendicular to the plane of the area is:

A. 0^0 B. 90^0 C. 45^0 D. 30^0

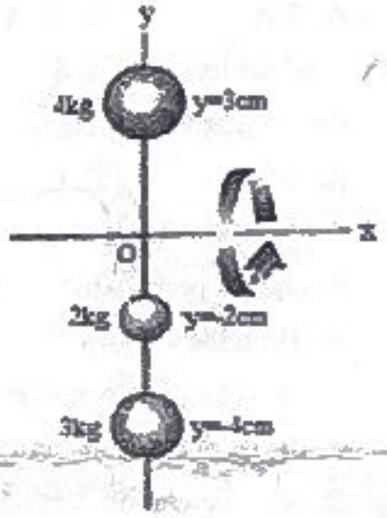
B 33. A long straight wire carrying a current produces a magnetic field of 0.8 T at a distance 0.5 cm from the wire. What is the magnetic field at a distance of 1 cm?

A. 0.2 T B. 0.4 T C. 1.6 T D. 3.2 T

D 34. The emf induced in a coil can be increased by:

A. increasing the number of turns per unit length in the coil (n).
B. increasing the rate of change of magnetic field strength surrounding the coil per unit time.
C. increasing the speed of the relative motion between the coil and the magnet.
D. All of the above are correct statements.

C 35. Three particles are connected by rigid rods of negligible mass lying along the *y*-axis as shown in figure below. If the system rotates about the *x*-axis with angular speed of 2 rad/s, what is the moment of inertia about the *x*-axis?



A. 36 kg.cm
B. 4 kg.cm
C. 92 kg.cm^2
D. 20 kg.cm^2

D 36. Which of the following is true about magnetic field lines?

A. They form closed loops.
B. They never intersect each other.
C. The magnetic field lines are crowded near the pole.
D. All are true statements.

B 37. A 500 turns coil develops an average induced voltage of 60 V. Over what time interval must a flux change of 0.06 Wb occur to produce such a voltage?

A. 5.0 sec B. 0.5 sec C. 2.0 sec D. 0.2 sec

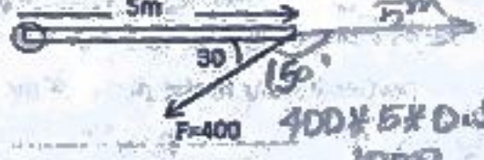
B 38. Which one of the following is false about magnets?

A. Every magnet has two poles.
B. Like poles attract each other.
C. A magnet generates a magnetic field.
D. The magnetic field is stronger at the poles.

B 39. Two objects are attracted to each other by a gravitational force F. If the distance between the objects is doubled, what is the new gravitational force between the objects in terms of F?

A. 4 F B. ¼ F C. 16 F D. 1/16 F

D 40. A force of 400 N is applied to a beam at a distance of 5 m from the pivot point, as shown in figure below. What is the magnitude of the torque which turns the bar around pivot?



A. 2000 Nm C. 867 Nm
B. 1733 Nm D. 1000 Nm

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Name: Hanna Ayalew **Grade &sec:** 12th C **Number:** 24

1. A	17. D	29. B
2. B	18. D	30. D
3. B	19. C	31. C
4. A	20. C	32. A
5. A	21. B	33. B
6. B	22. A	34. D
7. A	23. C	35. C
8. A	24. D	36. D
9. B	25. C	37. B
10. B	26. A	38. B
11. A	27. A	39. B
12. A	28. B	40. D
13. B		
14. C		
15. C		
16. B		

41. Pascal's Principle says that an increase in pressure in one part of an enclosed liquid result in an increase in pressure throughout all parts of that liquid. Why then does the pressure differ at different heights?

(*) pascal's principle works for pressure of points of a fluid that are observed in the same height of the container

$F = mg$ $P = F/A = \frac{mg}{A} = \rho gh$ → This means the pressure of a fluid at any point in a container depends on the height (Depth).

$A = V/h$

(*) $P_1 = P_2$

$\rho g h_1 = \rho g h_2$ If and only if $h_1 = h_2$ for $P_1 = P_2$

- so there is variation of pressure if there is variation of depth of observation.